



BSI Standards Publication

Conductors of insulated cables

National foreword

This British Standard is the UK implementation of EN IEC 60228:2024. It is identical to IEC 60228:2023. It supersedes [BS EN 60228:2005](#), which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee GEL/20/16, Electric Cables - Medium/high voltage.

A list of organizations represented on this committee can be obtained on request to its committee manager.

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Conductors of insulated cables
(IEC 60228:2023)

Ames des câbles isolés
(IEC 60228:2023)

Leiter für Kabel und isolierte Leitungen
(IEC 60228:2023)

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European Committee for Electrotechnical Standardization
Comité Européen de Normalisation Electrotechnique
Europäisches Komitee für Elektrotechnische Normung

CEN-CENELEC Management Centre: Rue de la Science 23, B-1040 Brussels

European foreword

The text of document 20/2125/FDIS, future edition 4 of IEC 60228, prepared by IEC/TC 20 "Electric cables" was submitted to the IEC-CENELEC parallel vote and approved by CENELEC as EN IEC 60228:2024.

The following dates are fixed:

- latest date by which the document has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 2025-03-12
- latest date by which the national standards conflicting with the document have to be withdrawn (dow) 2027-06-12

This document supersedes EN 60228:2005 and all of its amendments and corrigenda (if any).

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Endorsement notice

The text of the International Standard IEC 60228:2023 was approved by CENELEC as a European Standard without any modification.

Annex ZA
(normative)

**Normative references to international publications
with their corresponding European publications**

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE 1 Where an International Publication has been modified by common modifications, indicated by (mod), the relevant EN/HD applies.

NOTE 2 Up-to-date information on the latest versions of the European Standards listed in this annex is available here: www.cenelec.eu.

<u>Publication</u>	<u>Year</u>	<u>Title</u>	<u>EN/HD</u>	<u>Year</u>
IECEE OD-5014	-	IEC System of Conformity Assessment Schemes for Electrotechnical Equipment and Components (IECEE System), Committee of Testing Laboratories (CTL), Instrument Accuracy Limits	-	-

Annex ZB
(normative)

Special national conditions

Special national condition: National characteristic or practice that cannot be changed even over a long period, e. g. climatic conditions, electrical earthing conditions.

NOTE If it affects harmonization, it forms part of the European Standard or Harmonization Document.

For the countries in which the relevant special national conditions apply these provisions are normative, for other countries they are informative.

Clause Special national condition

Table 3 Cyprus, Ireland, United Kingdom

Add:

1,25 0,21 15,6 16,1

NOTE This conductor is for cables which are intended for use on appliances fitted with 13 A plugs conforming to BS 1363-1 or I.S. 401

Table C.1 Cyprus, Ireland, United Kingdom

Add:

1,25 – – 1,7

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

CONDUCTORS OF INSULATED CABLES

FOREWORD

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IEC 60228 has been prepared by IEC technical committee 20: Electric cables. It is an International Standard.

This fourth edition cancels and replaces the third edition published in 2004. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) a description of Milliken conductors has been added;
- b) nominal cross-sectional areas above 2 500 mm² have been added;

- c) the old 2 500 mm² aluminium resistance value has been corrected and a new value introduced.

For legacy systems where the 2 500 mm² aluminium conductor was designed taking into account the value presented in previous editions and no longer tabulated, then the original design can be maintained and still utilized.

The suppliers can furthermore utilize such superseded design of 2 500 mm² aluminium conductors either in systems already designed and qualified but not delivered or for example to produce repair and additional spare lengths for delivered systems.

The choice of utilizing the original superseded design of 2 500 mm² aluminium conductors or a new one based on the new resistance tabulated value is a matter of agreement between the supplier and final users.

The text of this International Standard is based on the following documents:

Draft	Report on voting
20/2125/FDIS	20/2131/RVD

Full information on the voting for its approval can be found in the report on voting indicated in the above table.

The language used for the development of this International Standard is English.

This document was drafted in accordance with ISO/IEC Directives, Part 2, and developed in accordance with ISO/IEC Directives, Part 1 and ISO/IEC Directives, IEC Supplement, available at www.iec.ch/members_experts/refdocs. The main document types developed by IEC are described in greater detail at www.iec.ch/publications.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under webstore.iec.ch in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn, or
- revised.

INTRODUCTION

This document is intended as a fundamental reference standard for IEC technical committees and National Committees in drafting standards for electric cables, and to the National Committees in drafting specifications for use in their own countries. These committees select from the tables of this general standard the conductors appropriate to the particular applications relevant to them and either include the applicable details in their cable specifications or make appropriate references to this document.

CONDUCTORS OF INSULATED CABLES

1 Scope

This document specifies the nominal cross-sectional areas, in the range 0,5 mm² to 3 500 mm², for conductors in electric power cables and cords of a wide range of types. Requirements for numbers and sizes of wires and resistance values are also included. These conductors include solid, stranded and Milliken, copper, aluminium and aluminium alloy conductors in cables for fixed installations and flexible copper conductors.

This document does not apply to conductors for telecommunication purposes.

The applicability of this document to a particular type of cable is as specified in the standard for the type of cable.

Unless specified otherwise in a particular clause, this document relates to the conductors in the finished cable and not to the conductor as made or supplied for inclusion into a cable.

Conductors described in this document are specified in metric sizes.

Informative annexes provide supplementary information covering temperature correction factors for resistance measurement (Annex B) and guidance on dimensional limits of circular conductors (Annex C).

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IECEE OD-5014, *Instrument Accuracy Limits*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>

3.1

metal-coated

coated with a thin layer of suitable metal, such as tin or tin alloy

3.2

nominal cross-sectional area

value that identifies a particular size of conductor but is not subject to direct measurement

Note 1 to entry: Each particular size of conductor in this document is required to meet a maximum resistance value.

3.3**Milliken conductor**

stranded conductor comprising an assembly of shaped conductors, lightly insulated from each other

4 Classification

The conductors have been divided into four classes, 1, 2, 5 and 6. Those in classes 1 and 2 are intended for use in cables for fixed installations. Classes 5 and 6 are intended for use in flexible cables and cords but can also be used for fixed installations.

- class 1: solid conductors;
- class 2: stranded conductors;
- class 5: flexible conductors;
- class 6: flexible conductors made with smaller diameter wires than class 5 for the same nominal conductor cross-section.

5 Materials**5.1 General**

The wires of conductors shall consist of one of the following (except for the Milliken central element):

- plain or metal-coated annealed copper;
- aluminium or aluminium alloy.

The wires of stranded conductors (for example Milliken conductors) can be oxidized or enamelled.

5.2 Circular and shaped solid aluminium conductors

Circular and shaped solid aluminium conductors shall be made from aluminium such that the tensile strength of the completed conductor is within the limits given in Table 1.

Table 1 – Tensile strength limits for circular and shaped solid aluminium conductors

Nominal cross-sectional area mm ²	Tensile strength N/mm ²
10 and 16	110 to 165
25 and 35	60 to 130
50	60 to 110
70 and above	60 to 90

There are no limits defined for the tensile strength of aluminium alloy solid conductors.

5.3 Circular and shaped stranded aluminium conductors

Stranded aluminium conductors shall be made from aluminium such that the tensile strength of the individual wires is within the limits given in Table 2:

Table 2 – Tensile strength limits for circular and shaped stranded aluminium conductors

Nominal cross-sectional area mm ²	Tensile strength N/mm ²
10	up to 200
16 and above	125 to 205

There are no limits defined for the tensile strength of aluminium alloy stranded conductors.

This data can only be checked on wires taken before stranding and not on wires taken from a stranded conductor.

6 Solid conductors and stranded conductors

6.1 Solid conductors (class 1)

6.1.1 Construction

- a) Solid conductors or conductor wires shall consist of one of the materials specified in Clause 5.
- b) Solid copper conductors shall be of circular cross-section.

NOTE Solid copper conductors having nominal cross-section areas of 25 mm² and above are for particular types of cable, e.g. mineral insulated, and not for general purposes.

- c) Solid aluminium and solid aluminium alloy conductors of sizes 10 mm² to 35 mm² shall be of circular cross-section. Larger sizes shall be of circular cross-section for single-core cables and may be of either circular or shaped cross-section for multi-core cables.

6.1.2 Resistance

The resistance of each conductor at 20 °C, when determined in accordance with Clause 8, shall not exceed the appropriate maximum value given in Table 3.

For solid aluminium alloy conductors, having the same nominal cross-sectional area as an aluminium conductor, the resistance value given in Table 3 should be multiplied by a factor of 1,162 unless otherwise agreed between the manufacturer and the purchaser.

6.2 Stranded circular non-compacted conductors (class 2)

6.2.1 Construction

- a) Stranded circular non-compacted conductors (class 2) shall consist of one of the materials specified in Clause 5.
- b) Stranded aluminium or aluminium alloy conductors shall have a cross-sectional area not less than 10 mm².
- c) The wires in each conductor shall all have the same nominal diameter.
- d) The number of wires in each conductor shall be not less than the appropriate minimum number given in Table 4.

6.2.2 Resistance

The resistance of each conductor at 20 °C, when determined in accordance with Clause 8, shall not exceed the appropriate maximum value given in Table 4.

6.3 Stranded compacted circular conductors and stranded shaped conductors (class 2)

6.3.1 Construction

- a) Stranded compacted circular conductors and stranded shaped conductors (class 2) shall consist of one of the materials specified in Clause 5. Stranded compacted circular aluminium or aluminium alloy conductors shall have a nominal cross-sectional area not less than 10 mm². Stranded compacted circular copper conductors shall have a nominal cross-sectional area not less than 1,5 mm². Stranded shaped copper, aluminium or aluminium alloy conductors shall have a nominal cross-sectional area of not less than 25 mm².
- b) The ratio of the diameters of two different wires in the same conductor shall not exceed 2, except for conductors made with pre-shaped wires.
- c) The number of wires in each conductor shall be not less than the appropriate minimum number given in Table 4, except for conductors made with pre-shaped wires. This requirement applies to conductors made with wires of circular cross-section before compaction.

6.3.2 Resistance

The resistance of each conductor at 20 °C, when determined in accordance with Clause 8, shall not exceed the appropriate maximum value given in Table 4.

6.4 Milliken conductors (class 2)

6.4.1 Construction

- a) Milliken conductors (class 2) shall consist of any of the materials specified in Clause 5. Milliken conductors shall have a nominal cross-sectional area not less than 800 mm², and the minimum number of wires for each cross section is not specified.
- b) The ratio of the diameters of two different wires of any segment shall not exceed 2 (Milliken central element excluded).
- c) The Milliken central element can be empty, with a solid conductor, wires or with a plastic filler.
- d) This conductor may be constructed from 4, 5, or 6 equal segments. The number of wires in each segment is defined by the manufacturer. In the case of a central element formed with wires, it can be considered as an additional segment with a polygonal shape.

6.4.2 Resistance

The resistance of the whole conductor at 20 °C, for all different constructions described in 6.4.1, when determined in accordance with Clause 8, shall not exceed the appropriate maximum value given in Table 4.

7 Flexible conductors (classes 5 and 6)

7.1 Construction

- a) Flexible conductors (classes 5 and 6) shall consist of plain or metal-coated annealed copper.
- b) The wires in each conductor shall have the same nominal diameter.
- c) The diameter of the wires in each conductor shall not exceed the appropriate maximum value given in Table 5 for class 5 or Table 6 for class 6 conductors.

7.2 Resistance

The resistance of each conductor at 20 °C, when determined in accordance with Clause 8, shall not exceed the appropriate maximum value given in Table 5 or Table 6.

8 Check of compliance with Clause 6 and Clause 7

Compliance with the requirements for construction of 6.1.1, 6.2.1, 6.3.1, 6.4.1 and 7.1 shall be checked on the completed cable by inspection and measurement where practicable.

Compliance with the requirements for resistance given in 6.1.2, 6.2.2, 6.3.2, 6.4.2, and 7.2 shall be checked by measurement in accordance with Annex A and corrected for temperature by the factors in Table A.1.

Table 3 – Class 1 solid conductors for single-core and multi-core cables

1	2	3	4
Nominal cross-sectional area	Maximum resistance of conductor at 20 °C		
	Circular, annealed copper conductors		Aluminium and aluminium alloy conductors, circular or shaped ^c
	Plain	Metal-coated	
mm ²	Ω/km	Ω/km	Ω/km
0,5	36,0	36,7	-
0,75	24,5	24,8	-
1,0	18,1	18,2	-
1,5	12,1	12,2	-
2,5	7,41	7,56	-
4	4,61	4,70	-
6	3,08	3,11	-
10	1,83	1,84	3,08 ^a
16	1,15	1,16	1,91 ^a
25	0,727 ^b	-	1,20 ^a
35	0,524 ^b	-	0,868 ^a
50	0,387 ^b	-	0,641
70	0,268 ^b	-	0,443
95	0,193 ^b	-	0,320 ^d
120	0,153 ^b	-	0,253 ^d
150	0,124 ^b	-	0,206 ^d
185	0,101 ^b	-	0,164 ^d
240	0,077 5 ^b	-	0,125 ^d
300	0,062 0 ^b	-	0,100 ^d
400	0,046 5 ^b	-	0,077 8
500	-	-	0,060 5
630	-	-	0,046 9
800	-	-	0,036 7
1 000	-	-	0,029 1
1 200	-	-	0,024 7
1 400	-	-	0,021 2
1 600	-	-	0,018 6

^a Aluminium conductors 10 mm² to 35 mm² circular only; see 6.1.1 c).

^b See note in 6.1.1 b).

^c See note in 6.1.2.

^d For single-core cables, four sectoral shaped conductors may be assembled into a single circular conductor. The maximum resistance of the assembled conductor shall be 25 % of that of the individual component conductors.

Table 4 – Class 2 stranded conductors for single-core and multi-core cables

1	2	3	4	5	6	7	8	9	10
Nominal cross-sectional area mm ²	Minimum number of wires in the conductor						Maximum resistance of conductor at 20 °C		
	Circular		Circular compacted		Shaped		Annealed copper conductor		Aluminium or aluminium alloy conductor ^b Ω/km
	Cu	Al	Cu	Al	Cu	Al	Plain wires Ω/km	Metal-coated wires Ω/km	
0,5	7	-	-	-	-	-	36,0	36,7	-
0,75	7	-	-	-	-	-	24,5	24,8	-
1,0	7	-	-	-	-	-	18,1	18,2	-
1,5	7	-	6	-	-	-	12,1	12,2	-
2,5	7	-	6	-	-	-	7,41	7,56	-
4	7	-	6	-	-	-	4,61	4,70	-
6	7	-	6	-	-	-	3,08	3,11	-
10	7	7	6	6	-	-	1,83	1,84	3,08
16	7	7	6	6	-	-	1,15	1,16	1,91
25	7	7	6	6	6	6	0,727	0,734	1,20
35	7	7	6	6	6	6	0,524	0,529	0,868
50	19	19	6	6	6	6	0,387	0,391	0,641
70	19	19	12	12	12	12	0,268	0,270	0,443
95	19	19	15	15	15	15	0,193	0,195	0,320
120	37	37	18	15	18	15	0,153	0,154	0,253
150	37	37	18	15	18	15	0,124	0,126	0,206
185	37	37	30	30	30	30	0,099 1	0,100	0,164
240	37	37	34	30	34	30	0,075 4	0,076 2	0,125
300	61	61	34	30	34	30	0,060 1	0,060 7	0,100
400	61	61	53	53	53	53	0,047 0	0,047 5	0,077 8
500	61	61	53	53	53	53	0,036 6	0,036 9	0,060 5
630	91	91	53	53	53	53	0,028 3	0,028 6	0,046 9
800	91	91	53	53	-	-	0,022 1	0,022 4	0,036 7
1 000	91	91	53	53	-	-	0,017 6	0,017 7	0,029 1
1 200	91	91	53	53	-	-	0,015 1	0,015 1	0,024 7
1 400 ^a	91	91	53	53	-	-	0,012 9	0,012 9	0,021 2
1 600	91	91	53	53	-	-	0,011 3	0,011 3	0,018 6
1 800 ^a	-						0,010 1	0,010 1	0,016 5
2 000	-						0,009 0	0,009 0	0,014 9
2 500	-						0,007 2	0,007 2	0,011 9
3 000 ^a	-						0,006 0	0,006 0	0,009 9
3 200	-						0,005 6	0,005 6	0,009 3
3 500 ^a	-						0,005 1	0,005 1	0,008 5

^a These sizes are non-preferred. Other non-preferred sizes are recognized for some specialized applications but are not within the scope of this document.

^b For stranded aluminium alloy conductors having the same nominal cross-sectional area as an aluminium conductor the resistance value should be agreed between the manufacturer and the purchaser.

Table 5 – Class 5 flexible copper conductors for single-core and multi-core cables

1	2	3	4
Nominal cross-sectional area mm ²	Maximum diameter of wires in conductor mm	Maximum resistance of conductor at 20 °C	
		Plain wires Ω/km	Metal-coated wires Ω/km
0,5	0,21	39,0	40,1
0,75	0,21	26,0	26,7
1,0	0,21	19,5	20,0
1,5	0,26	13,3	13,7
2,5	0,26	7,98	8,21
4	0,31	4,95	5,09
6	0,31	3,30	3,39
10	0,41	1,91	1,95
16	0,41	1,21	1,24
25	0,41	0,780	0,795
35	0,41	0,554	0,565
50	0,41	0,386	0,393
70	0,51	0,272	0,277
95	0,51	0,206	0,210
120	0,51	0,161	0,164
150	0,51	0,129	0,132
185	0,51	0,106	0,108
240	0,51	0,080 1	0,081 7
300	0,51	0,064 1	0,065 4
400	0,51	0,048 6	0,049 5
500	0,61	0,038 4	0,039 1
630	0,61	0,028 7	0,029 2

Table 6 – Class 6 flexible copper conductors for single-core and multi-core cables

1	2	3	4
Nominal cross-sectional area mm ²	Maximum diameter of wires in conductor mm	Maximum resistance of conductor at 20 °C	
		Plain wires Ω/km	Metal-coated wires Ω/km
0,5	0,16	39,0	40,1
0,75	0,16	26,0	26,7
1,0	0,16	19,5	20,0
1,5	0,16	13,3	13,7
2,5	0,16	7,98	8,21
4	0,16	4,95	5,09
6	0,21	3,30	3,39
10	0,21	1,91	1,95
16	0,21	1,21	1,24
25	0,21	0,780	0,795
35	0,21	0,554	0,565
50	0,31	0,386	0,393
70	0,31	0,272	0,277
95	0,31	0,206	0,210
120	0,31	0,161	0,164
150	0,31	0,129	0,132
185	0,41	0,106	0,108
240	0,41	0,080 1	0,081 7
300	0,41	0,064 1	0,065 4

Annex A (normative)

Measurement of resistance

The cable shall be kept in the test area for a sufficient time to ensure that the conductor temperature has reached a level which permits an accurate determination of resistance using the correction factors provided.

Measure the DC resistance of the conductor(s), either on a complete length of cable or flexible cord or on a sample of cable or flexible cord of at least 1 m in length, at room temperature and record the temperature at which the measurement is made. Adjust the measured resistance by means of the correction factors given in Table A.1.

Calculate the resistance per kilometre length of cable from the length of the complete cable and not from the length of the individual core or wires.

If necessary, correction to 20 °C and 1 km length shall be made by applying the following formula:

$$R_{20} = R_t \times k_t \times \frac{1000}{L}$$

where

k_t is the temperature correction factor from Table A.1;

R_{20} is the conductor resistance at 20 °C, in Ω/km ;

R_t is the measured conductor resistance, in Ω ;

L is the length of the cable, in m.

The expanded measurement uncertainty ($k = 2$) for R_{20} shall be in accordance with IEC 60228:2023 Annex D.

Table A.1 – Temperature correction factors k_t for conductor resistance to correct the measured resistance at t °C to 20 °C

1	2	1	2
Temperature of conductor at time of measurement t °C	Correction factor k_t All conductors	Temperature of conductor at time of measurement t °C	Correction factor k_t All conductors
0	1,087	21	0,996
1	1,082	22	0,992
2	1,078	23	0,988
3	1,073	24	0,984
4	1,068	25	0,980
5	1,064	26	0,977
6	1,059	27	0,973
7	1,055	28	0,969
8	1,050	29	0,965
9	1,046	30	0,962
10	1,042	31	0,958
11	1,037	32	0,954
12	1,033	33	0,951
13	1,029	34	0,947
14	1,025	35	0,943
15	1,020	36	0,940
16	1,016	37	0,936
17	1,012	38	0,933
18	1,008	39	0,929
19	1,004	40	0,926
20	1,000		
NOTE The values of correction factors k_t are based on a resistance-temperature coefficient of 0,004 per Kelvin at 20 °C. The values of temperature correction factors specified in column 2 are approximate but give practical values well within the accuracy that can normally be achieved in measurements of conductor temperature and length of cable or flexible cords. For more accurate values for the temperature correction factors for copper and aluminium, reference should be made to Annex B. However, these should not be treated as a requirement for testing in compliance with this document in the assessment of resistances.			

Annex B (informative)

Exact formulae for the temperature correction factors

a) Annealed copper conductors: plain or metal-coated

$$k_{t,Cu} = \frac{254,5}{234,5 + t} = \frac{1}{1 + 0,00393(t - 20)}$$

b) Aluminium conductors

$$k_{t,Al} = \frac{248}{228 + t} = \frac{1}{1 + 0,00403(t - 20)}$$

For aluminium alloys, reference should be made to the manufacturer.

In all the above cases, t refers to the temperature of the conductor at the time of measurement in degrees Celsius.

Annex C (informative)

Guidance on the dimensional limits of circular conductors

C.1 Purpose

This Annex C is intended as a guide for manufacturers of cables and cable connectors to assist in ensuring that the conductors and connectors are dimensionally compatible. It gives guidance on dimensional limits for the following types of conductor included in this document:

- a) circular solid conductors, (class 1) of copper, aluminium or aluminium alloy;
- b) circular and compacted circular stranded conductors, (class 2) of copper, aluminium or aluminium alloy;
- c) flexible conductors, (classes 5 and 6) of copper.

C.2 Dimensional limits for circular copper conductors

The diameters of circular copper conductors should not exceed the values given in Table C.1.

If minimum diameters for class 1 circular copper conductors are required, reference can be made to the minimum diameters for solid circular aluminium or aluminium alloy conductors indicated in Table C.3.

C.3 Dimensional limits for stranded compacted circular copper, aluminium and aluminium alloy conductors

The diameters of stranded compacted circular copper, aluminium and aluminium alloy conductors should not exceed the maximum values and should be not less than the minimum values given in Table C.2.

In the exceptional case of uncompacted circular stranded aluminium or aluminium alloy conductors, the maximum diameters should not exceed the corresponding values for copper conductors given in column 3 of Table C.1.

C.4 Dimensional limits for circular solid aluminium conductors

The diameters of circular solid aluminium and aluminium alloy conductors should not exceed the maximum values and should be not less than the minimum values given in Table C.3.

Table C.1 – Maximum diameters of solid, non-compacted stranded and flexible circular copper conductors

1	2	3	4
Nominal cross-sectional area mm ²	Conductors in cables for fixed installations		Flexible conductors (Classes 5 and 6) mm
	Solid (Class 1) mm	Stranded (Class 2) mm	
0,5	0,9	1,1	1,1
0,75	1,0	1,2	1,3
1,0	1,2	1,4	1,5
1,5	1,5	1,7	1,8
2,5	1,9	2,2	2,4
4	2,4	2,7	3,0
6	2,9	3,3	3,9
10	3,7	4,2	5,1
16	4,6	5,3	6,3
25 ^a	5,7	6,6	7,8
35 ^a	6,7	7,9	9,2
50 ^a	7,8	9,1	11,0
70 ^a	9,4	11,0	13,1
95 ^a	11,0	12,9	15,1
120 ^a	12,4	14,5	17,0
150 ^a	13,8	16,2	19,0
185	15,4	18,0	21,0
240	17,6	20,6	24,0
300	19,8	23,1	27,0
400	22,2	26,1	31,0
500	-	29,2	35,0
630	-	33,2	39,0
800	-	37,6	-
1 000	-	42,2	-
NOTE The values given for flexible conductors are intended to allow for both class 5 and class 6 conductors.			
^a See 6.1.1 b).			

Table C.2 – Minimum and maximum diameters of stranded compacted circular copper, aluminium and aluminium alloy conductors

1	2	3
Nominal cross-sectional area mm ²	Stranded compacted circular conductors (Class 2)	
	Minimum diameter mm	Maximum diameter mm
10	3,6	4,0
16	4,6	5,2
25	5,6	6,5
35	6,6	7,5
50	7,7	8,6
70	9,3	10,2
95	11,0	12,0
120	12,3	13,5
150	13,7	15,0
185	15,3	16,8
240	17,6	19,2
300	19,7	21,6
400	22,3	24,6
500	25,3	27,6
630	28,7	32,5
NOTE 1 The dimensional limits of aluminium conductors with cross-sectional areas above 630 mm ² are not given as the compaction technology is not generally established.		
NOTE 2 No values are given for compacted copper conductors in the size range 1,5 mm ² to 6 mm ² .		

Table C.3 – Minimum and maximum diameters of solid circular aluminium conductors

1	2	3
Nominal cross-sectional area mm ²	Solid conductors (Class 1)	
	Minimum diameter mm	Maximum diameter mm
10	3,4	3,7
16	4,1	4,6
25	5,2	5,7
35	6,1	6,7
50	7,2	7,8
70	8,7	9,4
95	10,3	11,0
120	11,6	12,4
150	12,9	13,8
185	14,5	15,4
240	16,7	17,6
300	18,8	19,8
400	21,2	22,2
500	24,0	25,1
630	27,3	28,4
800	30,9	32,1
1000	34,8	36,0
1200	37,8	39,0

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